Project Title Computational Drug Discovery: Building Machine Learning Regression Models for Dopamine D2 Receptor Bioactivity Prediction

1. Project Abstract

- **The discovery of novel therapeutic drugs is a time-consuming and resource-intensive process.**
- **This project leverages cheminformatics and machine learning to accelerate the early stages of drug discovery. The specific focus of this pipeline is the Dopamine D2 receptor.**
- **By extracting bioactivity data from the ChEMBL database, calculating molecular descriptors, and training regression models, this project will develop a computational tool capable of predicting the inhibitory effects of novel chemical compounds against the Dopamine D2 receptor.**

2. Introduction and Background

- **The Dopamine D2 receptor is a transmembrane protein. It is heavily implicated in various complex neuropsychiatric conditions.**
- **Inhibiting this receptor is a proven therapeutic strategy.**
- **However, testing millions of chemical compounds in a laboratory setting to find active inhibitors is highly inefficient.**
- **By standing at the intersection of chemistry and computer science, this project utilizes a data-driven approach.**
- **Instead of physical assays, machine learning algorithms will be trained on historical bioactivity data to recognize complex structural patterns in molecules.**
- **This computational screening acts as a powerful filter, prioritizing the most promising compounds for subsequent biological testing.**

3. Project Objectives

- **To programmatically retrieve and curate a high-quality dataset of chemical compounds and their corresponding bioactivity values against the Dopamine D2 receptor from the ChEMBL database.**

- **To perform exploratory data analysis evaluating the drug-likeness of active versus inactive compounds using Lipinski Rule of Five.**
- **To compute 1D and 2D molecular descriptors that numerically represent the chemical structure of each compound.**
- **To train, evaluate, and optimize a Random Forest regression model capable of predicting the exact pIC50 bioactivity values.**
- **To benchmark the Random Forest model against a suite of alternative regression algorithms to determine the most accurate predictive pipeline.**

4. Methodology

The project will be executed in five distinct phases, adapting established computational drug discovery workflows to the new protein target.

- **Phase 1: Data Collection and Preprocessing**
    - **Data will be retrieved using the ChEMBL web resource client in Python.**
    - **The target queried will be the human Dopamine D2 receptor, specifically filtering for IC50 standard values.**
    - **The raw dataset will be cleaned by handling missing data, standardizing chemical structures, and converting raw IC50 measurements into the negative logarithmic scale pIC50 to ensure a more normal distribution for regression modeling.**
- **Phase 2: Exploratory Data Analysis**
    - **Compounds will be categorized into active, inactive, and intermediate classes based on their bioactivity thresholds.**
    - **Molecular properties including molecular weight, logarithm of the partition coefficient, number of hydrogen bond donors, and number of hydrogen bond acceptors will be calculated.**
    - **Statistical tests, such as the Mann-Whitney U test, will be applied alongside scatter plots and box plots to identify statistically significant differences between active and inactive inhibitors. These**

box plots will be specifically leveraged to evaluate the interquartile range (IQR) and statistical bounds, enabling a precise identification of data outliers and a deeper analysis of the distributions across compound classes.

- **Phase 3: Descriptor Calculation**
  - **To translate chemical structures into a machine-readable format, molecular descriptors will be generated.**
  - **The simplified molecular-input line-entry system strings of each compound will be processed using tools like PaDEL-Descriptor to compute PubChem fingerprints, resulting in a robust feature matrix for model training.**
- **Phase 4: Model Building**
  - **A Random Forest regression algorithm will be implemented as the baseline predictive model.**
  - **The dataset will be split into training and testing subsets.**
  - **The model will learn the non-linear relationships between the molecular fingerprints and the pIC50 values.**
  - **Model performance will be evaluated using metrics such as the coefficient of determination and mean squared error.**
- **Phase 5: Algorithm Comparison and Reporting**
  - **To ensure optimal predictive performance, the project will expand beyond Random Forest by evaluating a variety of machine learning regressors.**
  - **Algorithms such as Support Vector Regression, Gradient Boosting, and Ridge Regression will be trained and compared.**
  - **LazyPredict or similar automation libraries will be utilized to generate a comprehensive performance leaderboard, identifying the superior algorithm for this specific chemical space. The final clean**

datasets and model metrics can subsequently be exported for advanced dashboarding and interactive reporting using Power BI.

5. Expected Outcomes

- **A fully functional, end-to-end Python pipeline for retrieving, processing, and modeling biochemical data.**
- **A highly accurate predictive model that can estimate the binding affinity of new molecular structures against the Dopamine D2 receptor.**
- **A comprehensive analytical report detailing the chemical features most strongly associated with successful receptor inhibition.**

6. Tools and Technologies

- **Programming Language: Python**
- **Data Manipulation: Pandas, NumPy**
- **Data Visualization & Reporting: Matplotlib, Seaborn, Power BI**
- **Cheminformatics Libraries: RDKit, ChEMBL Webresource Client, PaDEL-Descriptor**
- **Machine Learning: Scikit-learn, LazyPredict**